

Breaking substitution ciphers

Exhaustive search

ciphertext only

known plaintext

chosen plaintext

chosen ciphertext

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Doable with a computer

Statistical analysis

Match letters together

Choose ABCDE ... and match letters

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chosen ciphertext	Choose ABCDE ... and match letters

Polyalphabetic ciphers (a.k.a Renaissance Cipher)

➔ Vigenere cipher

Algorithm : combine the message and the key

Key : a word

Key space : 26^n (n being the length of the key)

wearediscoveredsaveyourself	
+ <u>deceptivedeceptivedeceptive</u>	(mod 26)
ZICVTWQNGRZGVTWAVZHCQYGLMGJ	

Advantage : Encryption of a letter is context dependent